Problem 1. (a) reflexive : $a \le a+1$ ✓

symmetric : $a \le b+1 \not\Rightarrow b \le a+1$: $a=1, b=3$ ✗

anti-symmetric : $a=3, b=4$: $a \ne b$, $a \le b+1$, $b \le a+1$. ✗

transitive : $a \le b+1$, $b \le c+1 \not\Rightarrow a \le c+1$: $a=6, b=5, c=4$ ✗.

(b) reflexive : $\gcd(a,a) = a > 1$ if $a \ge 2$. ✗

symmetric : $\gcd(a,b) = \gcd(b,a)$ ✓

anti-symmetric : ✗

transitive : $a=2, b=3, c=4$ $\gcd(a,b) = \gcd(b,c) = 1$

but $\gcd(a,c) = 2$ ✗

(c) reflexive : $a=2$ ✗

symmetric : ✓

anti-symmetric : ✗

transitive : $a=c=1$, $b=2$. ✗

(d) reflexive : ✓

symmetric : x $(1,1)$, $(2,2)$

antisymmetric: ✓

transitive: ✓

Problem 2

$$\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & \multicolumn{4}{c}{} \\ \end{array}$$

$$\begin{array}{cc} & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \end{array}$$

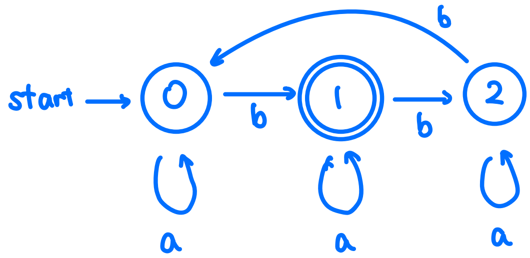
M is symmetric $\Leftrightarrow$ R is symmetric.

diagonal entries $=1 \Leftrightarrow$ R is reflexive

transitive : brute force.

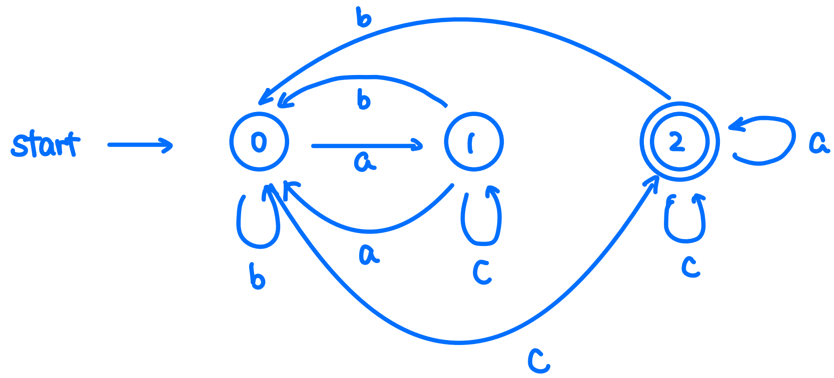
Equivalent classes : $\{1,2\}$, $\{3,4\}$.

Problem 3. (a)



accepted $\iff$ $\#b \equiv 1 \pmod 3$

(b)



accepted $\iff$ the last 'b' appears before the last 'c'

& there are even number of 'a's in between.

$$\cdots b \underbrace{\underbrace{\cdots}_{\#a \equiv 0 \pmod 2}\ c \cdots}_{b\text{-free}}$$

Problem 4. (i) $S = \{0,1\}$, $A = \{a,b\}$, $F = \{0\}$.

μ	a	b
0	0	1
1	1	1

(ii) $S = \{0,1,2,3,4,5\}$, $A = \{0,1,\cdots,9\}$, $F = \{4\}$.